B) [scan 25]

†

†

†

Since γ_2 vanishes at † P_ω, γ_3 at the Q_i, it follows that there is no invariant section of an $\mathcal{O}(n)^*$ † over Σ_J^* † for $n \leq 6$, hence that the order of $\mathrm{cl}(\mathcal{O}(1))$ in $H^1(\Sigma_J, \mathfrak{S}_J; \mathbb{G}_m)$ is indeed 6, which proves that $\mathcal{O}(1)$ is a generator of the mixed H^1. 24